# ROBUSTNESS

# Z SCORE RECAP

# Z SCORE

- z-score describes the location of the raw score
  - in terms of distance from the mean, measured in standard deviations
- Gives us information about the location of that score relative to the "average" deviation of all scores
- A z-score is the number of standard deviations a score is above or below the mean of the scores in a distribution.
- A raw score is a regular score before it has been converted into a Z score
- Raw scores on very different variables can be converted into Z scores and directly compared

# Z SCORE

Raw scores on very different variables can be converted into Z scores and directly compared

What does a z-score tell us

- Mean of zero
- Zero distance from the mean
- Standard deviation of 1
- The z-score has two parts
  - The number
  - The sign

# CONVERTING SCORES INTO Z SCORES AND BACK

- Now we will consider several examples that illustrate the use of the z-score formula. Suppose that we know about a population of a particular breed of cats having weights that are normally distributed. Furthermore, suppose we know that the mean of distribution is 10 pounds and the standard deviation is 2 pounds. Consider the following questions:
- What is the *z*-score for 13 pounds?
- What is the *z*-score for 6 pounds?
- How many pounds corresponds to a *z*-score of 1.25?

# CONVERTING SCORES INTO Z SCORES AND BACK

- What is the z-score for 13 pounds?
  - (13 – 10)/2 = 1.5
  - This means that 13 is one and a half standard deviations above the mean.
- What is the z-score for 6 pounds?
  - (6 – 10)/2 = -2
  - The interpretation of this is that 6 is two standard deviations below the mean.
- How many pounds corresponds to a *z*-score of 1.25?
  - $1.25 = (x - 10)/2$
  - $2.5 = (x - 10)$
  - $12.5 = x$
  - And so we see that 12.5 pounds corresponds to a *z*-score of 1.25.

# CONVERTING ACROSS MEASURES

- We can convert from one metric to another.
- National results for the SAT test show that for college-bound seniors the average combined SAT Writing, Math and Verbal score is 1500 and the standard deviation is 250.
- National results for the ACT test show that for college-bound seniors the average composite ACT score is 20.8 and the standard deviation is 4.8.
  - Your SAT score: 1860.
  - Your neighbor's ACT: 29.
- Who did better on their respective test, and how would each have done on the other test?

# CONVERTING ACROSS MEASURES

- Who did better on their respective test?
- SAT into Z
  - 1860-1500=360
  - 360/250=1.44
- ACT into Z
  - 29- 20.8=8.2
  - 8.2/ 4.8= 1.708

# CONVERTING ACROSS MEASURES

- How would each have done on the other test?
- ZSAT into ACT
  - 1.44*4.8=6.912
  - 6.912+20.8=27.712
- ZACT into SAT
  - 1.708*250=427
  - 427+1500=1927

# COMMENTS ON STANDARDIZATION

- Standardization of variables
  - Transformation – ratio variable
    - Features of measurement scales
    - Some have built in properties
      - Percentiles (not equal interval!)
      - Standardized scale (z score scale)

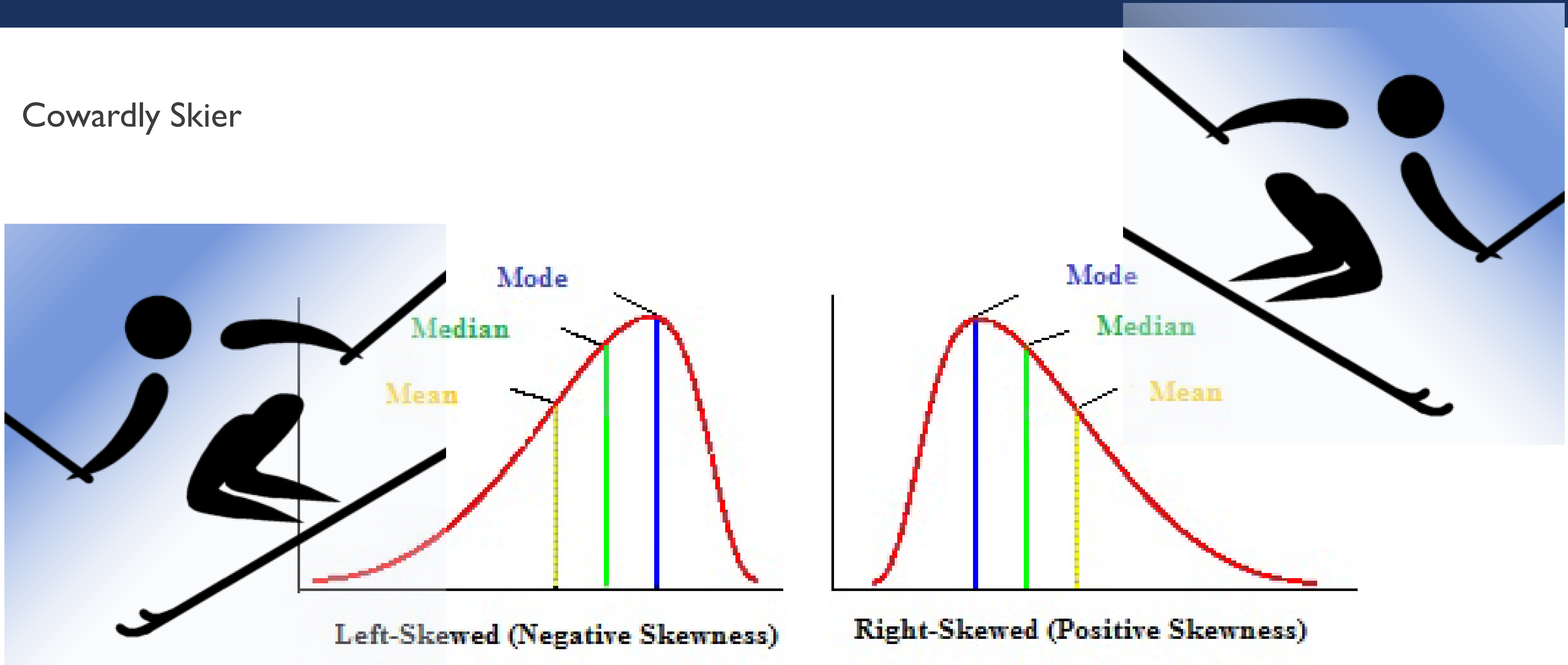
# PERCENTILES

- Percentiles (percentile rank)
- Score value that cuts off a certain % of scores below it (cumulative percentile)
- Special percentile ranks
  - M: 50%
  - Q1: 25%
  - Q3: 75%
- Don't have equal intervals
  - Weak statistically

# SKEWNESS

# SKEWNESS

- Cowardly Skier



# POSITIVE SKEW

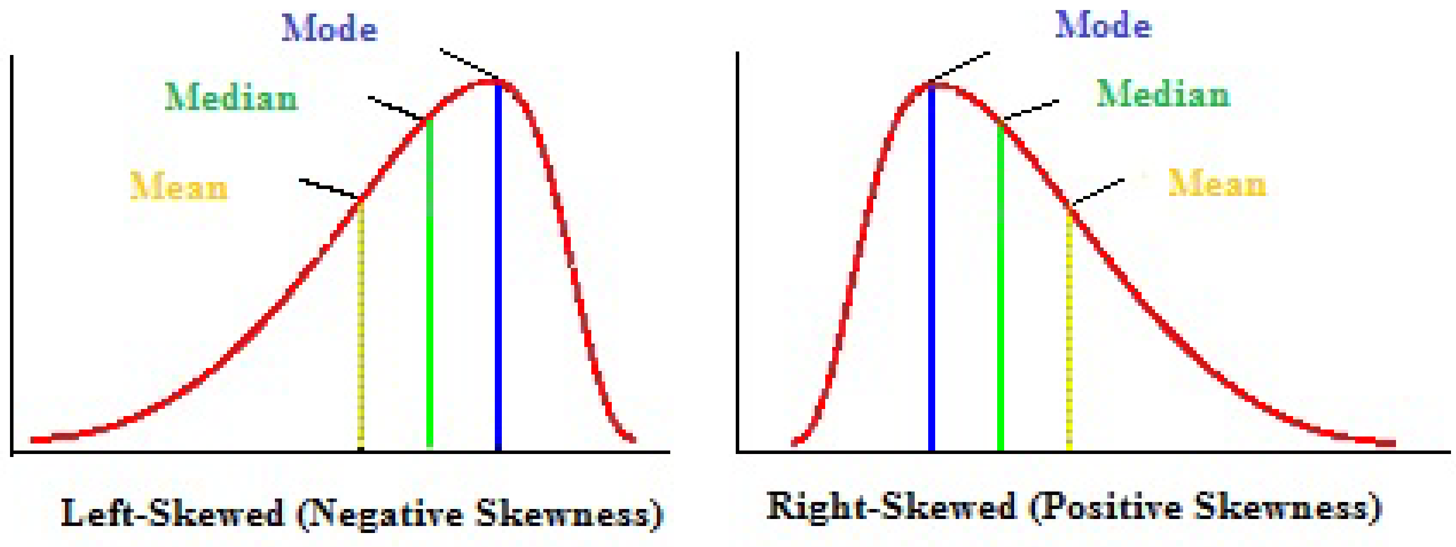
- The distribution is said to be *right-skewed*, *right-tailed*, or *skewed to the right*,
  - *despite* the fact that the curve itself appears to be skewed or leaning to the left;
- The right tail is longer;
  - the mass of the distribution is concentrated on the left
- *right* instead refers to the right tail being drawn out and, often,
- A right-skewed distribution usually appears as a *left-leaning* curve.

# NEGATIVE SKEW FACTS

- Also called: *left-skewed*, *left-tailed*, or *skewed to the left*
- The left tail is longer;
- the mass of the distribution is concentrated on the right of the figure.
- *left* instead refers to the left tail being drawn out
- A left-skewed distribution usually appears as a *right-leaning* curve.

# RELATIONSHIP BETWEEN MEAN, MEDIAN, AND MODE

- Symmetric unimodal distribution
  - Mean=median=mode



# WORKED EXAMPLE

- Where do students go to school?
  - Although 80.4% of first-time first-year students attended college in the state in which they lived, this percent varied considerably over the states.
  - Here is a stemplot of the percent of first-year students in each of the 50 states who were from the state where they enrolled.
  - The stems are 10s and the leaves are 1s.
  - The stems have been split in the plot.

| | |
|---|---|
| 3 | 4 |
| 3 | 8 |
| 4 | 0 |
| 4 | 6 |
| 5 | 0 |
| 5 | 6 |
| 6 | 0 0 1 2 2 3 4 |
| 6 | 5 9 |
| 7 | 2 2 2 4 4 4 |
| 7 | 5 6 6 6 6 7 7 8 8 9 |
| 8 | 0 0 1 2 2 3 |
| 8 | 5 5 7 7 8 9 |
| 9 | 0 0 1 2 3 3 4 |

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  - The stems are 10s and the leaves are 1s.
  - The stems have been split in the plot.
  - Questions
    - Where is the median?
    - The shape of the distribution is _____ skewed
    - The state with the smallest percent of first-year students enrolled in the state has what %
    - The state with the largest percent has %?

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|---|---|
| 3 | 4 |
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# WORKED EXAMPLE

- Questions
  - Where is the median?
  - The shape of the distribution is _____ skewed
  - The state with the smallest percent of first-year students enrolled in the state has v
  - The state with the largest percent has %?

| Stem | Leaves |
| --- | --- |
| 3 | 4 |
| 3 | 8 |
| 4 | 0 |
| 4 | 6 |
| 5 | 0 |
| 5 | 6 |
| 6 | 0 0 1 2 2 3 4 |
| 6 | 5 9 |
| 7 | 2 2 2 4 4 4 |
| 7 | 5 6 6 6 6 7 7 8 8 9 |
| 8 | 0 0 1 2 2 3 |
| 8 | 5 5 7 7 8 9 |
| 9 | 0 0 1 2 3 3 4 |

# WORKED EXAMPLE

- Questions
  - Where is the median?
    - 76
  - The shape of the distribution is left skewed
  - The state with the smallest percent of first-year students enrolled in the state has v
    - 34% enrolled.
  - The state with the largest percent has %?
    - 94%

| | |
|---|---|
| 3 | 4 |
| 3 | 8 |
| 4 | 0 |
| 4 | 6 |
| 5 | 0 |
| 5 | 6 |
| 6 | 0 0 1 2 2 3 4 |
| 6 | 5 9 |
| 7 | 2 2 2 4 4 4 |
| 7 | 5 6 6 6 6 7 7 8 8 9 |
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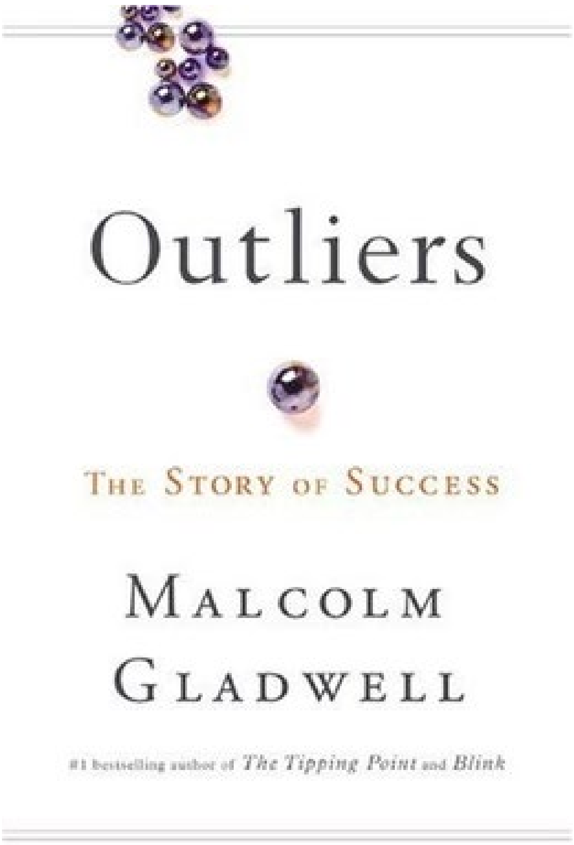


# ROBUST STATISTICS

# OUTLIERS

*The Story of Success*

MALCOLM GLADWELL

INTRODUCTION

## *The Roseto Mystery*

"THESE PEOPLE WERE DYING OF OLD AGE. THAT'S IT."

**out·li·er** \-ˌlī(-ə)r\ *noun*
**1:** something that is situated away from or classed differently from a main or related body
**2:** a statistical observation that is markedly different in value from the others of the sample

*1.*

# ROBUST STATISTICS AND OUTLIERS

- out-li-er noun
- 1: something that is situated away from or classed differently from a main or related body
- 2: a statistical observation that is markedly different in value from the others of the sample

- "An important kind of deviation is an **outlier**, an individual value that falls outside the overall pattern."

INTRODUCTION

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**out·li·er** \-ˌlī(-ə)r\ *noun*
**1:** something that is situated away from or classed differently from a main or related body
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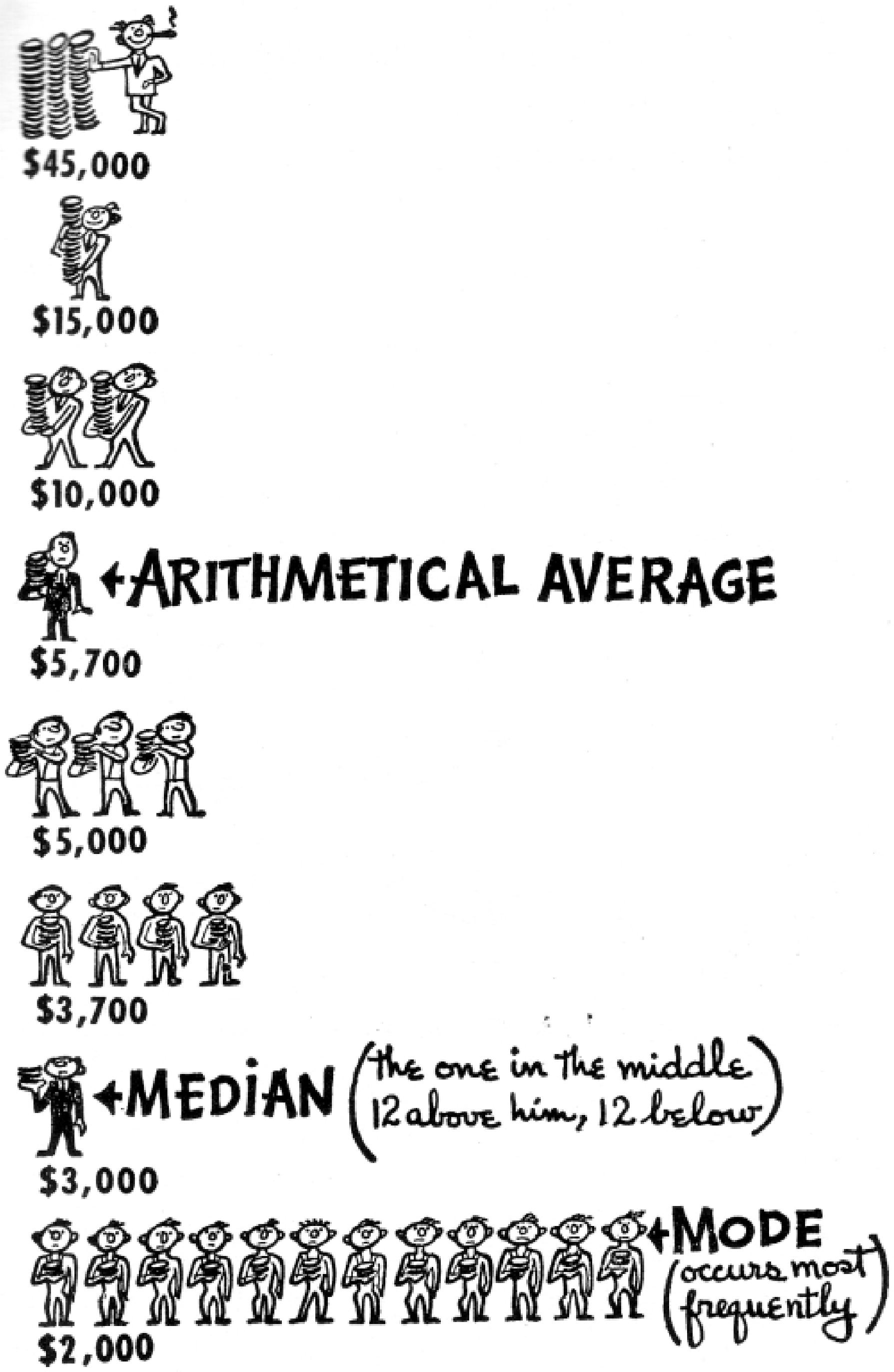
*1.*

# ROBUSTNESS

- Robust Statistics are less sensitive to outliers
- Most statistics, like
  - means, standard deviations, and correlations, and
  - every statistic based on these, are highly sensitive to outliers.



# SO WHAT DO WE DO WITH OUTLIERS?

Do we remove them?

Do we ignore them?

# EXAMPLE

- The data shows Tukey's scores for the last 5 math tests: 88, 90, 55, 94, and 89.
- Identify the outlier in the data set.
- Then determine how the outlier affects the mean, median, and mode of the data

# CALCULATIONS: MEAN

## With Outlier

- Mean: 55+88+89+90+94 = 416
  - 416 / 5 = 83.2
  - The mean is 83.2

## Without

- Mean: 88+89+90+94 = 361
  - 361 / 4 = 90.25
  - The mean is 90.25.

# CALCULATIONS: MEDIAN

## With Outlier

- Median: 55,88,<u>89</u>,90,94
  - The median is 89

## Without

- Median: 88,<u>89,90</u>,94
  - (89+90) / 2
  - The median is 89.5

# SO WHAT DO WE DO WITH OUTLIERS?

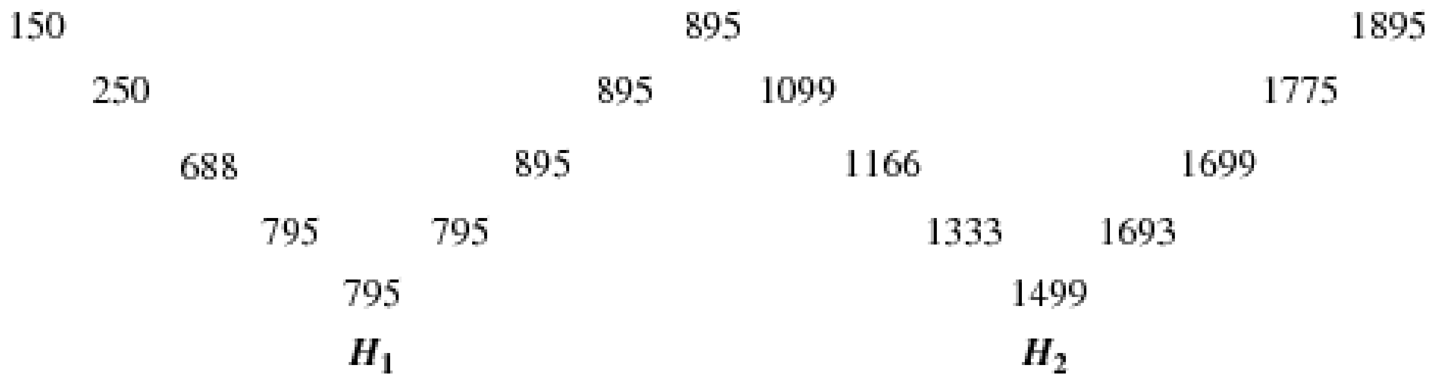
- Do we remove them?
- Do we ignore them?
  - It's up to the researcher

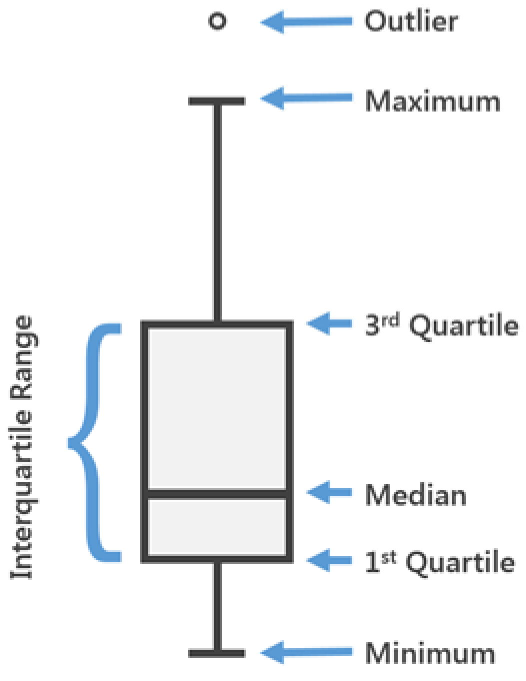
# SO WHAT DO WE DO WITH OUTLIERS?

- Do we remove them?
  - If you remove them, you need to document it
  - Why did you remove them?
    - Obvious error? Impossible number?
- Do we ignore them?
  - Easier, but it might alter our results

# 5- NUMBER SUMMARY TABLE

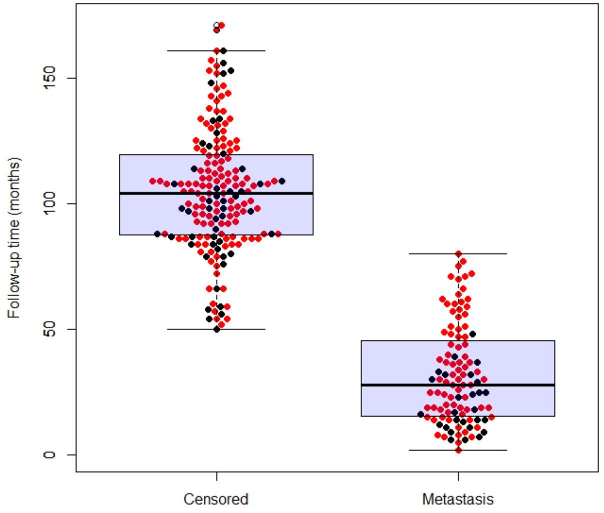
- Summarize a distribution
- Old (mean, standard deviation)
- Tukey's (lower extreme, lower hinge, median, upper hinge, upper extreme)
  - Commonly referred to (min, q1, m, q3, max)

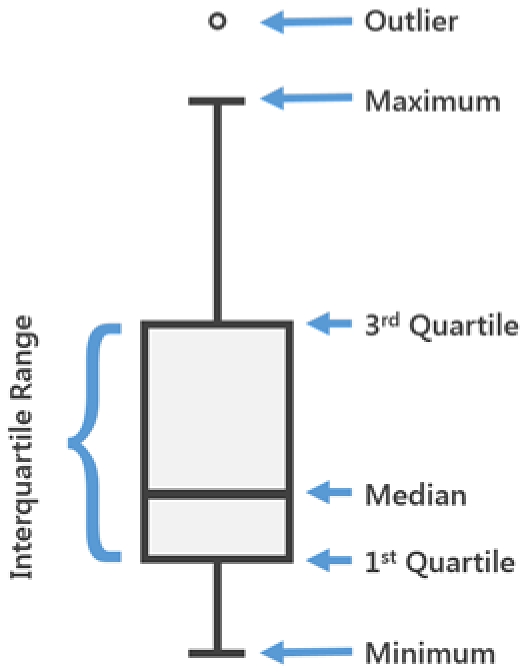


# 5- NUMBER SUMMARY TABLE

- 5- number summary table
  - Summarize a distribution
  - Old (mean, standard deviation)
  - Tukey's (lower extreme, lower hinge, median, upper hinge, upper extreme)
    - Commonly referred to (min, q1, m, q3, max)
  - Graph of the 5-summary table called boxplot
    - (lots of variants on this plot, including
      - Beeswarm Boxplot

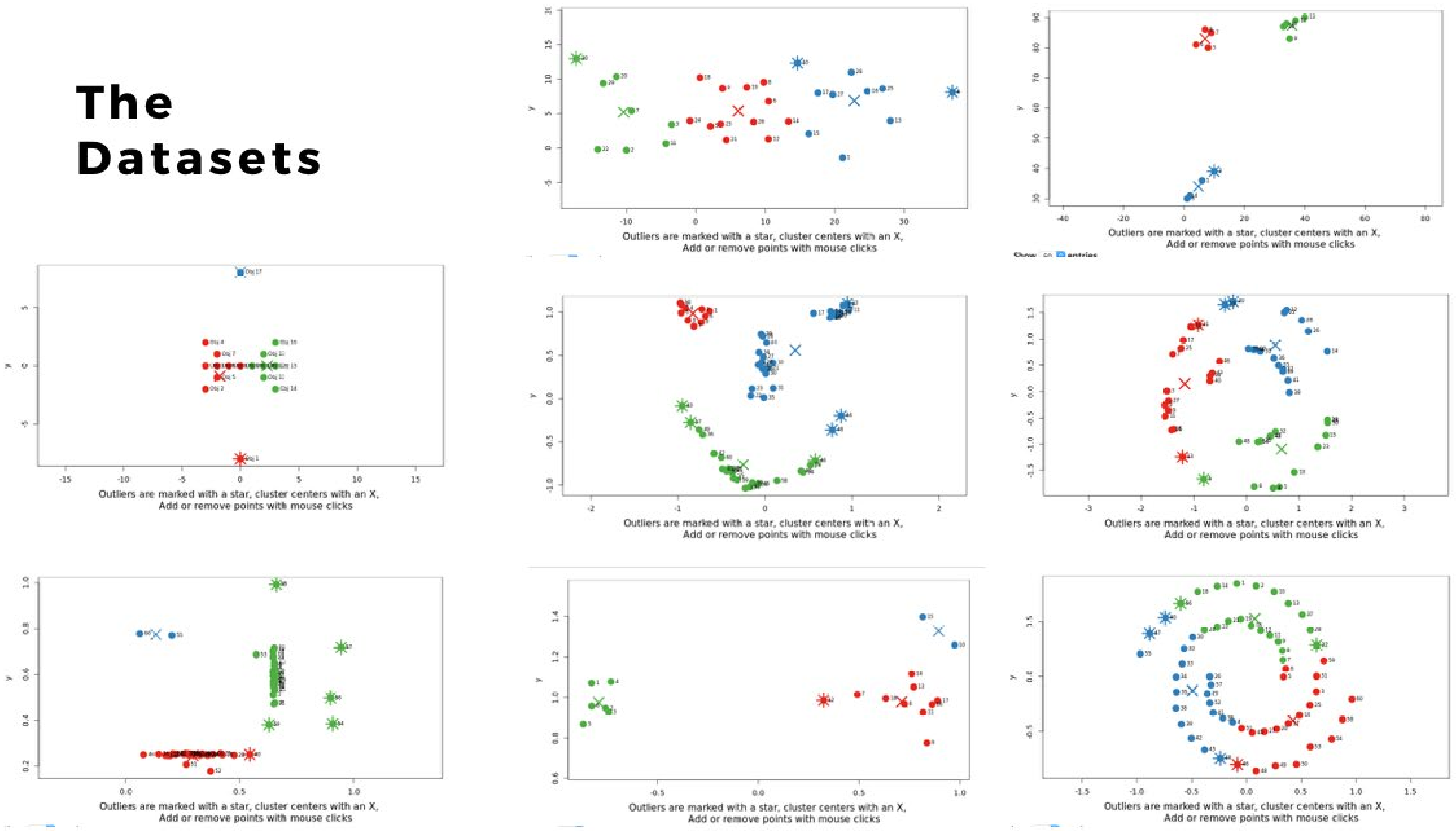


# DIAGNOSING OUTLIERS

- IQR rule (IQR= Interquartile Range (Q3-Q1)
  - Q3+ 1.5*IQR
  - Q1 - 1.5*IQR
  - Creates the fences outside of which the items
- Other Rules
  - 2.5 Standard Deviations beyond the mean
  - More Complex Algorithms
    - Illustration with Rshiny

# The Datasets

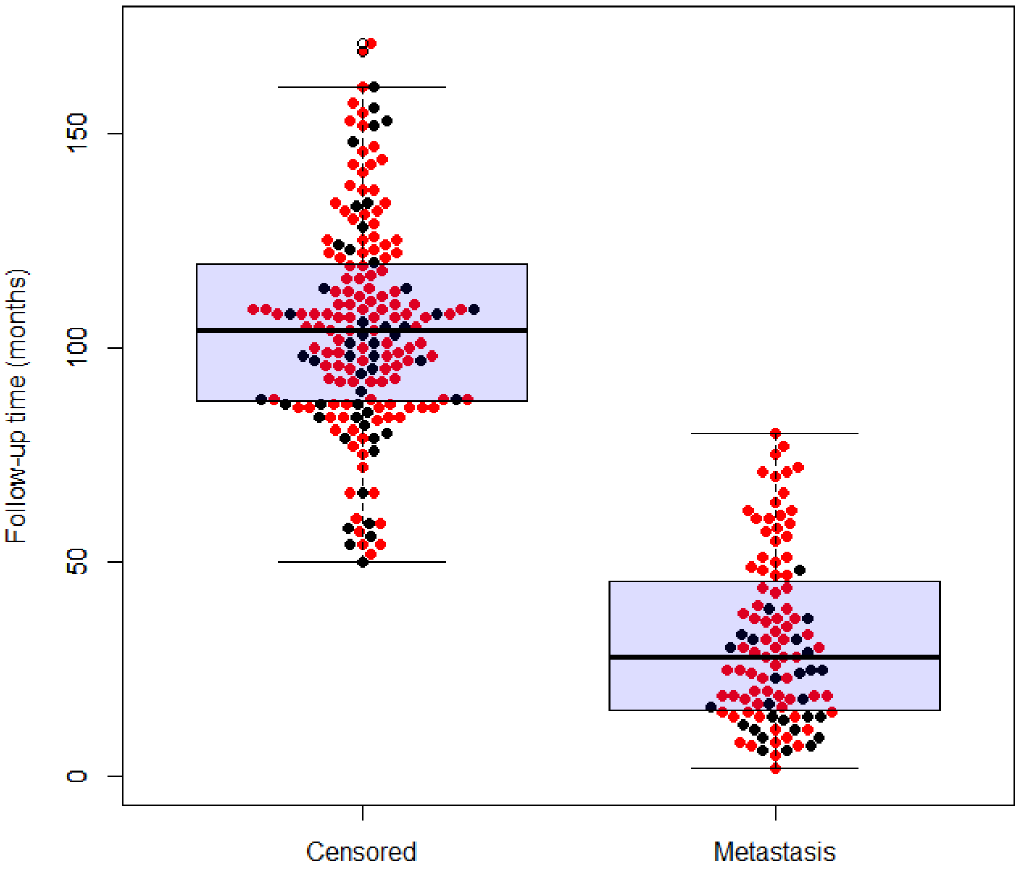


# WORKED EXAMPLE IN R

```r
library(dslabs)
library(dplyr)
library(ggplot2)

# These data are self-reported heights
data(reported_heights)
str(reported_heights)

# Here we create a new variable within the reported heights data
reported_heights <- reported_heights %>%
mutate(original_heights = height, height = as.numeric(height))

# Here we exclude heights with missing data
```

# R CODE FOR BEESWARM

```
if(!require(beeswarm)) install.packages(beeswarm)
data(breast)
beeswarm(time_survival ~ event_survival,
         data = breast, method = 'swarm', pch = 16,
         pwcol = as.numeric(ER), xlab = '',
         ylab = 'Follow-up time (months)',
         labels = c('Censored', 'Metastasis'))

boxplot(time_survival ~ event_survival, data = breast, add =
T, names = c("",""), col="#0000ff22")
```




https://www.r-statistics.com/2011/03/beeswarm-boxplot-and-plotting-it-with-r